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Capstone Presentation

Title Slide: Capstone Presentation

Credit Risk Prediction Using Machine Learning

Prepared by: Rohan Watave

Business Problem Understanding

Improve credit risk models for better loan approval and reduced default rates.

Challenges: Balancing risk and opportunity in lending decisions.

Solution: Use machine learning to enhance predictive accuracy.

Data Overview & Challenges

~100,000 customer records, 35+ features
Target variable: default (0 = No, 1 = Yes)

Class imbalance:

~78% non-default, 22% default

Financial variables are skewed, with many outliers

Modeling Approach Models used:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- AdaBoost

Why Ensemble Models?

- Handle non-linearity well
- Reduce overfitting via bootstrapping & boosting

Note:

- Random Forest had **train ROC AUC of 0.99** and **test ROC AUC of 0.8909**.

Model Performance Comparison

- Gradient Boosting and Random Forest generalize best

Model	Train ROC AUC	Test ROC AUC
Logistic Regression	0.89	0.87
Decision Tree	0.85	0.60
Random Forest	**0.99**	**0.89**
Gradient Boosting	**0.98**	**0.8886**
AdaBoost	0.93	0.86

Feature Importance

Top Predictors:

- worst_status_max (most delinquent account history)
- repayment_ratio (paid vs owed amount)
- recovery_debt (debt pending recovery)

Business Insight:

- Customers with high worst_status or low repayment are much more likely to default.

Key Insights from Modeling

- Ensemble models (Random Forest, Gradient Boosting) outperform traditional ones
- Class imbalance handled using **SMOTE**
- Complex relationships captured via boosted models

Business Recommendations

- 🚩 **Flag** customers with repayment_ratio < 0.5 for review
- ❌ Avoid loans to customers with worst_status_max >= 4
- 📈 Promote financial literacy programs for **young borrowers (< 30 yrs)**
- 🧠 Use Gradient Boosting for real-time credit scoring
- 🔄 Set up periodic retraining and monitoring

Next Steps:

- Integrate model into onboarding systems
- Conduct A/B testing on high-risk flagging logic
- Monitor ROC AUC monthly to detect drift
- Update training dataset every 6 months

Thank you